

# VAUGHN BERHOFF

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## EDUCATION

**Georgia Institute of Technology** - BS in Mechanical Engineering and minor in Industrial Design

Atlanta, GA

GPA: 3.63/4.00, Dean's List: Fall 2024, Spring 2025

May 2028 (Expected)

**Relevant Coursework:** Computing For Engineers, Statics, Engineering Graphics, Physics I, Physics II

## PROFESSIONAL EXPERIENCE

### Shurhold Industries

Palm City, FL

Factory Engineer Intern

May 2025 - Present

- Engineered a brush bumper installation mechanism, decreasing assembly time by 50%: <https://tinyurl.com/3mufvm8z>
- Improved upon existing processes, including automating a bucket lid assembly, reducing manual labor
- Generated technical drawings in SolidWorks to support documentation and reverse engineering of legacy parts
- Coordinated with vendors: machine shop, plastics manufacturer, and welding facility to source custom components

### Lineage Biomedical

Stuart, FL

Prototyping Intern

April 2024 - June 2024

- Collaborated with the lead engineer to develop functional prototypes for a new medical syringe to improve Botox injections
- Used SolidWorks to model components for analysis, reverse engineering, and design documentation

### Bikes Plus

Palm City, FL

Lead Mechanic

2020 - 2023

- Repaired, maintained, and built bikes of all types, including their drivetrain, suspension, braking systems, and custom fitting
- Gained strong customer service skills, respecting individual needs, resulting in increased returning customer rates
- Oversaw daily shop operations, including processing sales, facilitating repairs, cataloging inventory, and training new hires

## LEADERSHIP

### Georgia Tech. Invention Studio

May 2025 - Present

Prototyping Instructor (PI)

- Trained across 7+ fabrication areas: Metal Shop, CNC, Waterjet, Wood Shop, Laser Cutting, 3D Printing, and Electronics
- Supervise and support students users in safe tool operation, material selection, and DFM strategies

### Theme Park Engineering and Design club (TPED)

February 2025 - Present

Cruise Line Task Group Volunteer

- Volunteered at American Society for Testing and Materials (ASTM) F24 Amusement Rides and Devices conference
- Assisted Disney Standards Manager in synthesizing USTOA, ETOA, and Insurance guidelines for cruise excursion safety
- Case study on ICON Park Free Fall ride failure, identifying mechanical oversights and contributing safety recommendations
- Gained hands-on insight into regulatory processes, consensus-driven drafting, and real-world safety engineering practices

### Outdoor Recreation Georgia Tech. (ORGT)

January 2025 - Present

Whitewater Kayaking Instructor

- Structured custom lessons and challenges to teach new kayakers skills and confidence on the water
- Co-facilitated safety briefings and logistical planning for off-campus weekend trips, leading groups of 6 down river

## PROJECTS

### BoneYard - <https://youtu.be/2SjGUhyXE9I>

Drafting, Cardboard Models, Solidworks

- Designed and built a spooky-themed coin-activated pinball machine with a functional multi-ball and tri-flipper system
- Utilized rapid prototyping to develop 2D sketch to 3D model with over 90% recycled cardboard construction
- Integrated mechanical linkages, themed obstacles, and game flow logic to create an interactive experience

### Casino Watch

Fusion360, 3D Printer

- Stripped the movement and crown from a watch, inserting it into my newly 3D printed watch case, face, and hands
- Integrated mechanical and aesthetic elements through precision CAD modeling and part tolerancing

## SKILLS

**Tools & Software:** MATLAB, Solidworks, Laser Cutting, Wood & Metal fabrication, Waterjet, 3D Printing (FDM/SLA), CNC Machining, Soldering, Vinyl Cutting, Hand/Power Tools, cardboard prototyping

**Engineering Concepts:** CAD Modeling, Technical Drawing, Product R&D, Assembly lines, Tolerances, Materials Selection

**Soft skills:** Communication, Teamwork, Positivity, Peer Instruction, Leadership, Project Planning, Customer Interaction, Safety

**Certification:** SolidWorks (CSWA Certified), Fusion 360 certification, Advanced Wilderness First Aid (AWFA)